

Jude Vijayanga Wijesekera, MSc.

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 <https://judevijayanga.github.io/>

Dedicated researcher with 3+ years of expertise focused on galaxy formation and evolution in the early universe ($z \sim 6$). Demonstrated success in executing complex data pipelines and SED fitting using CIGALE to analyze the star-forming main sequence, infrared luminosity functions, and dust attenuation across cosmic time. Proven leader with experience heading academic staff at a national institute, combining advanced technical proficiency in Python (NumPy, Astropy) with a track record of high-impact publications.

Education

- October 2022 – September 2026  **PhD in Astronomy (reading)** Faculty of Physics, Astronomy and Informatics, Nicolaus Copernicus University, Poland.
Supervisor: dr. hab. Maciej Koprowski, Prof. UMK
Dissertation title: *Time evolution of the star formation rate density in the early Universe.*
- September 2020 – July 2022  **M.Sc. in Astrophysics and Cosmology** University of Padova, Italy.
Supervisor: Professor Paolo Cassata.
Thesis title: *Dust and molecular gas properties of a large sample of main-sequence galaxies at $3 < z < 4$.*
- July 2012 – February 2017  **BSc in Natural Sciences (Physics)** The Open University of Sri Lanka.
Thesis title: *Analysis of type II and type III solar radio burst using e-Calisto system.*

Work History

- 2018 – 2020  **Head of the Academic Staff**, Institute of Astronomy, Sri Lanka.
- Trained and mentored undergraduate students, overseeing their academic progress and development.
 - Designed, developed, and managed comprehensive course materials for the institute's educational programs.
 - Organized and directed practical observing camps to provide students with hands-on astronomical experience.
- 2017 – 2019  **Teaching Assistant**, Department of Physics, The Open University of Sri Lanka.
- Guided over 200 undergraduate students through complex practical laboratory sessions across a wide range of disciplines, including General Physics, Thermodynamics, Electromagnetism, Electronics, Optics, Waves, and Geology.
 - Supervised practical experiments to ensure safe and effective execution of laboratory protocols.
 - Evaluated student performance through the rigorous grading of coursework, lab reports, and the invigilation of examinations.

Other Research Experiences

28th July 2025- 1st August 2025	■ Internship at National Centre for Nuclear Research (NCBJ), Poland , carried out SIMBA cosmological simulation.
26th May 2025- 31st May 2025	■ Internship at National Centre for Nuclear Research (NCBJ), Poland , carried out SIMBA cosmological simulation.
June 2021-September 2021	■ Internship at International School for Advanced Studies (SISSA), Italy , Title: A Comprehensive X-ray view of the active nucleus in NGC 4258.
April 2019-April 2020	■ A project on probing emission-line properties of galaxies in the local Universe , Supervisor: Dr Themiya Nanayakkara, Postdoctoral Fellow, Swinburne University, Australia.

Research Publications

Peer-Reviewed Journals

- 1 M. P. Koprowski, K. Lisiecki, P. Sawant, and J. V. Wijesekera, "Stellar mass growth in COSMOS-Web: A mass-complete main sequence to $z \sim 8$ and its consistency with GSMF evolution," *Astronomy & Astrophysics*, 2026, Submitted to Astronomy & Astrophysics; manuscript available on Zenodo.  DOI: 10.5281/zenodo.21648421, submitted.
- 2 M. Koprowski et al., "Evolution of the infrared luminosity function and its corresponding dust-obscured star formation rate density out to $z \sim 6$," *Astronomy & Astrophysics*, vol. 706, A345, 2026.  DOI: 10.1051/0004-6361/202557505.
- 3 J. Wijesekera et al., "Evolution of dust attenuation in star-forming galaxies with uv slope, stellar mass, and redshift out to $z \sim 5$," *Astronomy & Astrophysics*, vol. 711, A189, 2026.  DOI: 10.1051/0004-6361/202659329.
- 4 M. Koprowski et al., "Charting the main sequence of star-forming galaxies out to redshifts $z \leq 5.7$," *Astronomy & Astrophysics*, vol. 691, A164, 2024.  DOI: 10.1051/0004-6361/202449948.
- 5 A. Masini, J. Wijesekera, A. Celotti, and P. G. Boorman, "Comprehensive x-ray view of the active nucleus in ngc 4258," *Astronomy & Astrophysics*, vol. 663, A87, 2022.  DOI: 10.1051/0004-6361/202243231.

Conference Proceedings

- 1 B. Perera et al., "Qualitative review of the photometric data of six ultra-faint dwarf galaxies," in *Proceedings of the International Conference on Applied and Pure Sciences (ICAPS 2021-Kelaniya)*, Faculty of Science, University of Kelaniya, Sri Lanka, vol. 1, 2021, p. 52.
- 2 B. Perera et al., "Stellar populations and the structural properties of six ultra-faint dwarf galaxies," in *Proceeding of the Open University Research Sessions (OURS 2021)*, The Open University of Sri Lanka, 2021, p. 147.
- 3 J. Wijesekera, K. Jayaratne, and J. Adassuriya, "Analysis of type ii and type iii solar radio bursts," in *Journal of Physics: Conference Series*, IOP Publishing, vol. 1005, 2018, p. 012 046.  DOI: 10.1088/1742-6596/1005/1/012046.
- 4 Y. Ramawickrama, J. Vijayanga, R. Dharmarathne, and H. Wijesooriya, "The future of gnss in the next ten years," in *Proceedings of the 3rd Asia Pacific Space Generation Workshop (3rd AP-SGW)*, Los Barnos, Philippines, 2016, pp. 12-13.

Books and Chapters

- 1 J. V. Wijesekera, Ed., *Practice for Astronomy Olympiad Book*. S&T Group, Sri Lanka, 2016, ISBN: 978-955-1080-24-2.

General Seminars and Talks

- 15 June 2026  ***“The Assembly of Star-Forming Galaxies: Evolution of the Main Sequence and Dust Obscuration out to $z \sim 6$ ”***
Place: Institute of Astronomy, Nicolaus Copernicus University, Toruń, Poland.

Skills

Languages	 Strong reading, writing and speaking competencies in English
Coding	 Python-advanced , Visual Basic, C++, Git hub, $\LaTeX$, ...
Astronomy packages/software	 CIGALE, MAGPHYS, DS9, Topcat, CASA, Fast++, XSpec, CIAO 4.13, FSPS.
Astronomy Databases & Surveys	 CANDELS fields, COSMOS, COSMOS-web & UDS catalogs, ALMA, SDSS (DR12 & DR16), GAIA survey, Chandra.
Web Dev	 HTML, css, JavaScript.
Misc.	 Academic research, teaching, $\LaTeX$ typesetting and publishing.

Miscellaneous Experience

Grants & Fellowships

- 2022-2026  **PhD Scholarship Grant**, Polish National Science Centre Sonata 16 Grant nr 2020/39/D/ST9/03078.
- 2020-2022  **Veneto Regional Scholarship**, University of Padova, Italy.

Awards

- 2017  **SPECTRUM Award for Excellence in Physics for Fundamentals of Geophysics**, The Open University of Sri Lanka.
- 2016  **SPECTRUM Award for Excellence in Physics for ‘Essentials of Geology**, The Open University of Sri Lanka.
-  **SPECTRUM Award for Excellence in Physics for Astronomy**, The Open University of Sri Lanka.
- 2011  **National Color Award for International Olympiad of Astronomy & Astrophysics**, University of Colombo, Sri Lanka.

Certifications

- 2024  **Fundamentals of Deep Learning**. Awarded by NVIDIA.
- 2023  **Scientific Writing and Publishing**. Awarded by Nature Masterclass.

Workshops & Schools

- 2025 - July  **Advanced Summer School Galaxy Formation and Evolution**, AKSS, Spetses (Greece).
- 2022  **11th IRAM Interferometry School**, Grenoble, France.
- 2018  **NARIT COMputational Astrophysics and Cosmology (NARIT COMAC) Workshop**, Chiang Mai, Thailand.
-  **NARIT-Institut d’Optique Graduate School. Optical Design Summer School**, Chiang Mai, Thailand.
- 2016  **Astronomical Data Reduction Workshop**, Arthur C Clarke Institute for Modern Technologies, Moratuwa, Sri Lanka.